

C++ Parking Fee Calculator — Code Summary

Overview

This program calculates a parking fee based on the number of hours entered by the user. It handles invalid input, applies a flat rate for short stays, and computes an extended fee for longer durations.

Fee Structure

- 0 or negative hours: Invalid input
- 1 to 2 hours: Flat fee of 100
- More than 2 hours: 100 base fee + 50 per extra hour beyond 2

How It Works

The `main()` function prompts the user to enter the number of hours. Based on the input, it either displays an error, prints the flat fee, or calls the `add()` function.

The `add()` function receives the hours as a parameter, calculates the extra hours beyond 2, multiplies them by 50, adds the base fee of 100, and prints the total.

Function: `add(int c)`

Takes the number of hours (`c`) as an integer parameter. Computes the total fee using the formula:

```
Total Fee = 100 + (c - 2) x 50
```

Example

Input: 5 hours

Extra hours = $5 - 2 = 3$

Total fee = $100 + (3 \times 50) = 250$